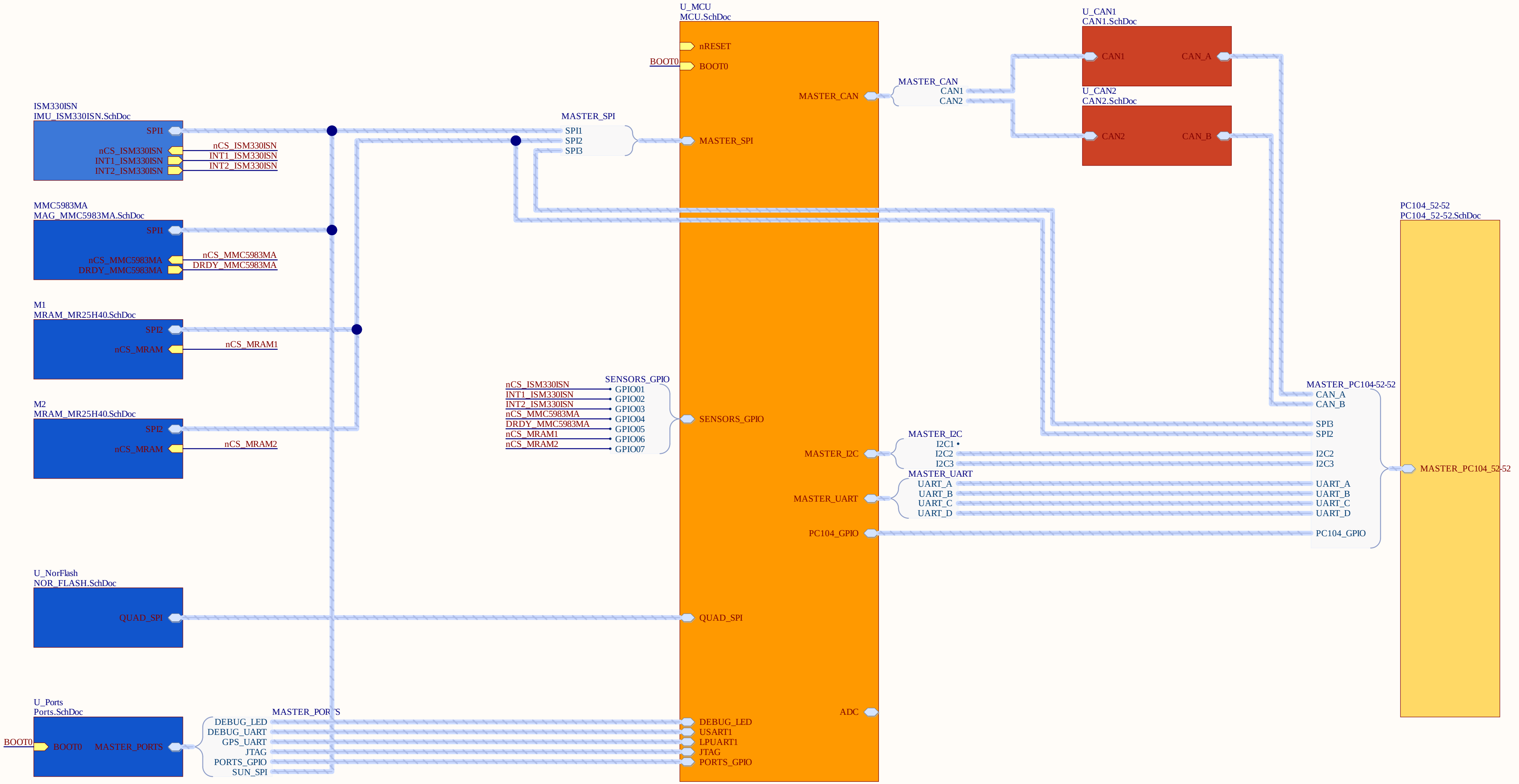
REVISION	DESCRIPTION	DATE	APPROVED


U_MCU
MCU.SchDoc
nRESET
BOOT0
BOOT0
MASTER_CAN
MASTER_SPI
SENSORS_GPIO
MASTER_I2C
MASTER_UART
PC104_GPIO
QUAD_SPI
ADC
DEBUG_LED
USART1
LPUART1
JTAG
PORTS_GPIO
U_CAN1
CAN1.SchDoc
CAN1
CAN_A
U_CAN2
CAN2.SchDoc
CAN2
CAN_B
MASTER_CAN
CAN1
CAN2
ISM330ISN
IMU_ISM330ISN.SchDoc
SPI1
nCS_ISM330ISN
INT1_ISM330ISN
INT2_ISM330ISN
nCS_ISM330ISN
INT1_ISM330ISN
INT2_ISM330ISN
MASTER_SPI
SPI1
SPI2
SPI3
MMC5983MA
MAG_MMC5983MA.SchDoc
SPI1
nCS_MMC5983MA
DRDY_MMC5983MA
nCS_MMC5983MA
DRDY_MMC5983MA
M1
MRAM_MR25H40.SchDoc
SPI2
nCS_MRAM
nCS_MRAM1
M2
MRAM_MR25H40.SchDoc
SPI2
nCS_MRAM
nCS_MRAM2
SENSORS_GPIO
nCS_ISM330ISN
INT1_ISM330ISN
INT2_ISM330ISN
nCS_MMC5983MA
DRDY_MMC5983MA
nCS_MRAM1
nCS_MRAM2
GPIO01
GPIO02
GPIO03
GPIO04
GPIO05
GPIO06
GPIO07
MASTER_I2C
I2C1
I2C2
I2C3
MASTER_UART
UART_A
UART_B
UART_C
UART_D
PC104_52-52
PC104_52-52.SchDoc
MASTER_PC104-52-52
CAN_A
CAN_B
SPI3
SPI2
I2C2
I2C3
UART_A
UART_B
UART_C
UART_D
PC104_GPIO
MASTER_PC104_52-52
U_NorFlash
NOR_FLASH.SchDoc
QUAD_SPI
U_Ports
Ports.SchDoc
BOOT0
BOOT0
MASTER_PORTS
MASTER_PORTS
DEBUG_LED
DEBUG_UART
GPS_UART
JTAG
PORTS_GPIO
SUN_SPI

APPROVALS	DATE
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CHK: *	

REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:

PROJECT
Altium™
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PROJECT REVISION: 4a74de86abeed4c235a77 DOCUMENT REVISION: DESIGN ITEM: [illegible]2c0ea51df00fe3c1041

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SIZE A3 CAGE CODE DWG NO. REV

SCALE: FILE NAME Top.SchDoc SHEET 1 OF 13

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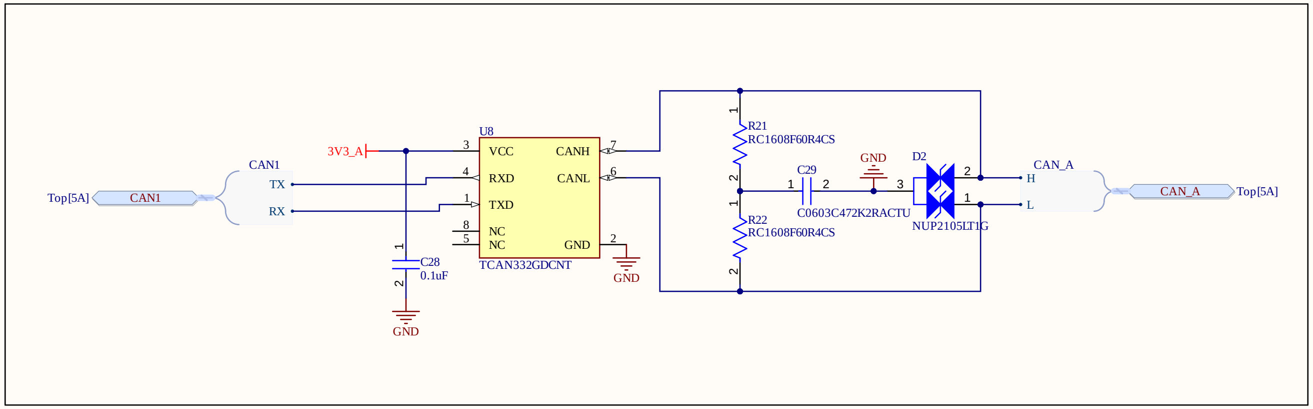
Top[5A]
CAN1
CAN1
TX
RX
3V3_A
U8
3
VCC
4
RXD
1
TXD
8
NC
5
NC
CANH
7
CANL
6
GND
2
TCAN332GDCNT
GND
1
C28
0.1uF
2
GND
1
R21
RC1608F60R4CS
2
1
R22
RC1608F60R4CS
2
C29
1
2
C0603C472K2RACTU
GND
3
D2
2
1
NUP2105LT1G
CAN_A
H
L
CAN_A
Top[5A]

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CHK: *	

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FAB DWG:

PCB DWG:

PROJECT

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DOCUMENT REVISION: [illegible]

DESIGN ITEM: [illegible]

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SIZE A3

CAGE CODE

DWG NO.

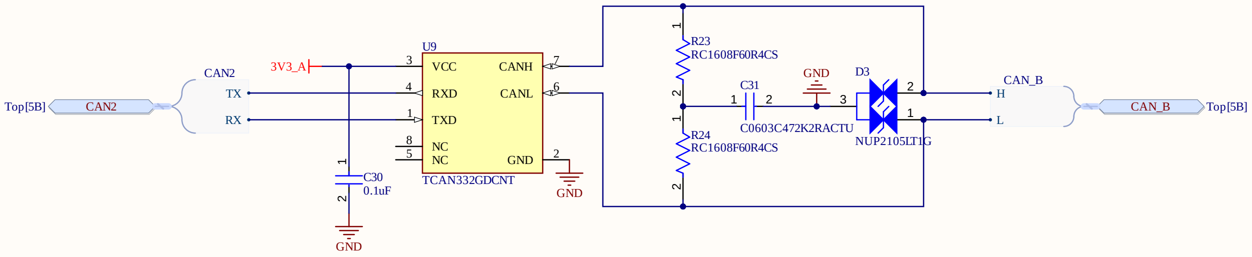
REV

SCALE:

FILE NAME CAN1.SchDoc

SHEET 2 OF 13

REVISION	DESCRIPTION	DATE	APPROVED


Top[5B]
CAN2
CAN2
TX
RX
3V3_A
U9
3
VCC
4
RXD
1
TXD
8
NC
5
NC
CANH
7
CANL
6
GND
2
TCAN332GDCNT
GND
1
C30
0.1uF
2
GND
1
R23
RC1608F60R4CS
2
1
R24
RC1608F60R4CS
2
C31
1
2
C0603C472K2RACTU
GND
3
D3
2
1
NUP2105LT1G
CAN_B
H
L
CAN_B
Top[5B]

APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	

REFERENCE DOCUMENTS

BOM:

ASSY DWG:

FAB DWG:

PCB DWG:

PROJECT

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PROJECT REVISION: 4a74de86abeed4c235a77[illegible]d500a6a357f9c92bd6

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SCALE:

FILE NAME CAN2.SchDoc

SHEET 3 OF 13

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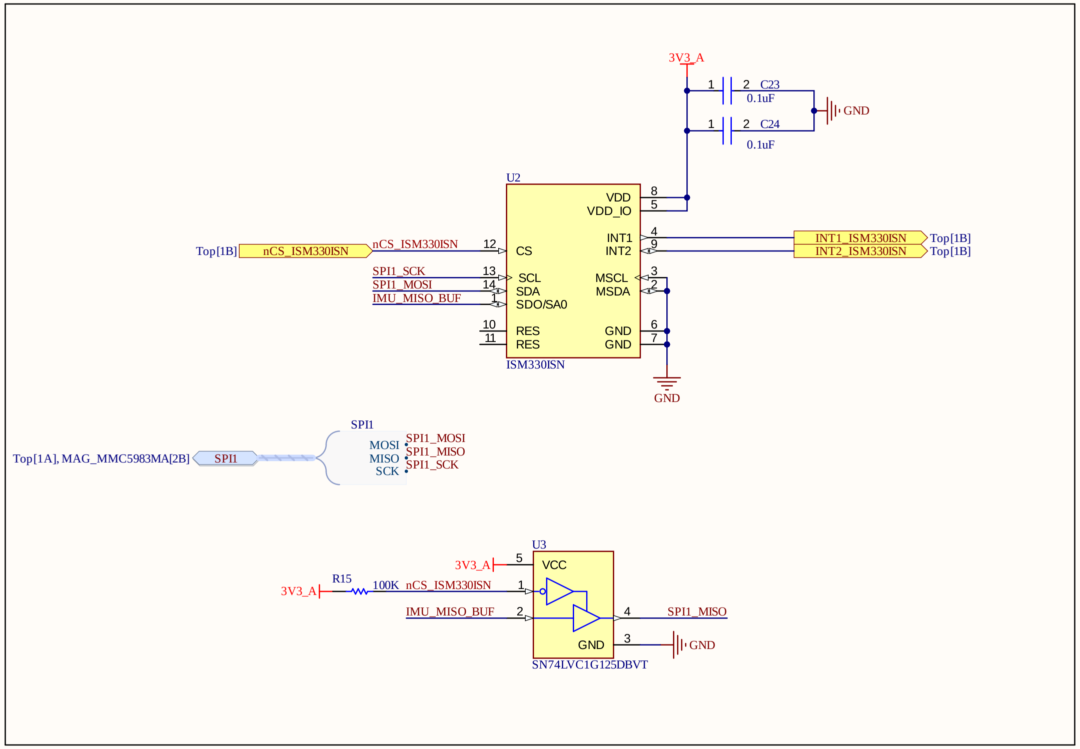
3V3_A
1
2
C23
0.1uF
GND
1
2
C24
0.1uF
U2
VDD
VDD_IO
8
5
INT1
INT2
4
9
INT1_ISM330ISN
INT2_ISM330ISN
Top[1B]
Top[1B]
Top[1B]
nCS_ISM330ISN
nCS_ISM330ISN
12
CS
SPI1_SCK
SPI1_MOSI
IMU_MISO_BUF
13
14
1
SCL
SDA
SDO/SA0
MSCL
MSDA
3
2
10
11
RES
RES
GND
GND
6
7
ISM330ISN
GND
SPI1
Top[1A], MAG_MMC5983MA[2B]
SPI1
MOSI
MISO
SCK
SPI1_MOSI
SPI1_MISO
SPI1_SCK
U3
3V3_A
5
VCC
3V3_A
R15
100K
nCS_ISM330ISN
1
IMU_MISO_BUF
2
4
SPI1_MISO
3
GND
GND
SN74LVC1G125DBVT

APPROVALS	DATE
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REFERENCE DOCUMENTS
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ASSY DWG:
FAB DWG:
PCB DWG:

PROJECT
Altium™
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PROJECT REVISION: 4a74de86abeed4c235a77[illegible]
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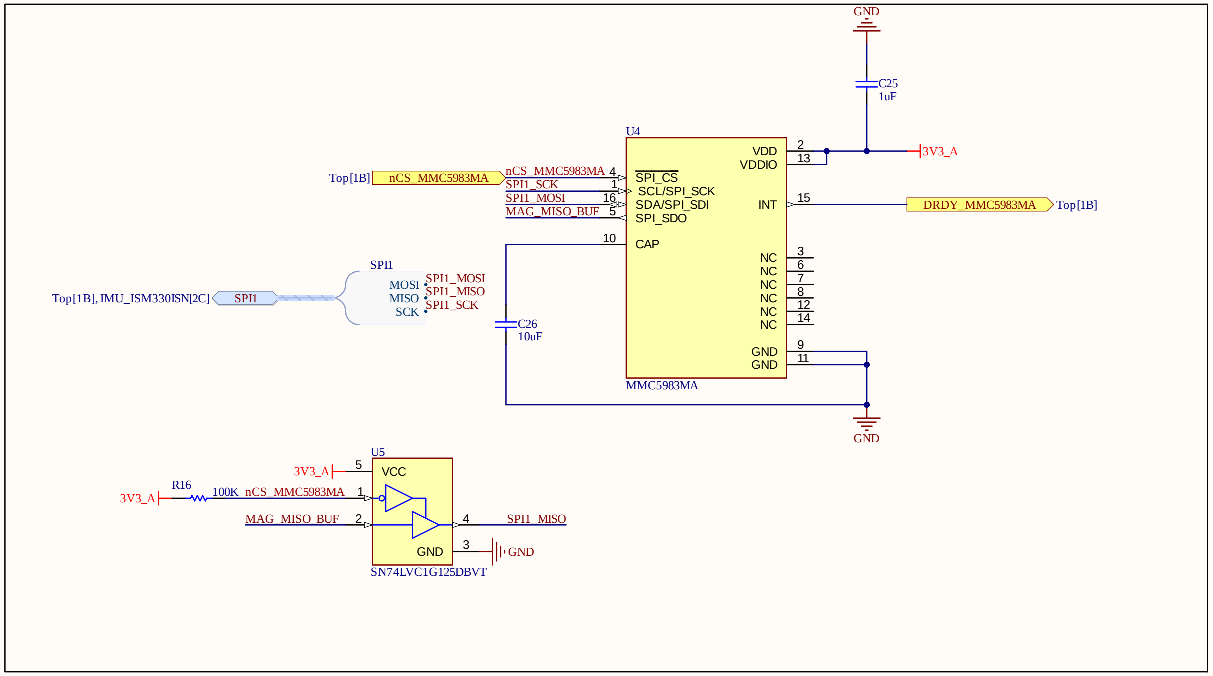
SHEET 4 OF 13

REVISION	DESCRIPTION	DATE	APPROVED


GND
C25
1uF
U4
VDD
VDDIO
2
13
3V3_A
Top[1B]
nCS_MMC5983MA
nCS_MMC5983MA
4
SPI1_SCK
1
SPI1_MOSI
16
MAG_MISO_BUF
5
SPI_CS
SCL/SPI_SCK
SDA/SPI_SDI
SPI_SDO
INT
15
DRDY_MMC5983MA
Top[1B]
10
CAP
NC
NC
NC
NC
NC
NC
3
6
7
8
12
14
SPI1
Top[1B], IMU_ISM330ISN[2C]
SPI1
MOSI
MISO
SCK
SPI1_MOSI
SPI1_MISO
SPI1_SCK
C26
10uF
9
11
GND
GND
MMC5983MA
GND
U5
3V3_A
5
VCC
R16
3V3_A
100K
nCS_MMC5983MA
1
MAG_MISO_BUF
2
4
SPI1_MISO
GND
3
GND
SN74LVC1G125DBVT

APPROVALS	DATE
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FAB DWG:
PCB DWG:

PROJECT
Altium™
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PROJECT REVISION: 4a74de86abeed4c235a77[illegible]
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SIZE A3
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DWG NO.
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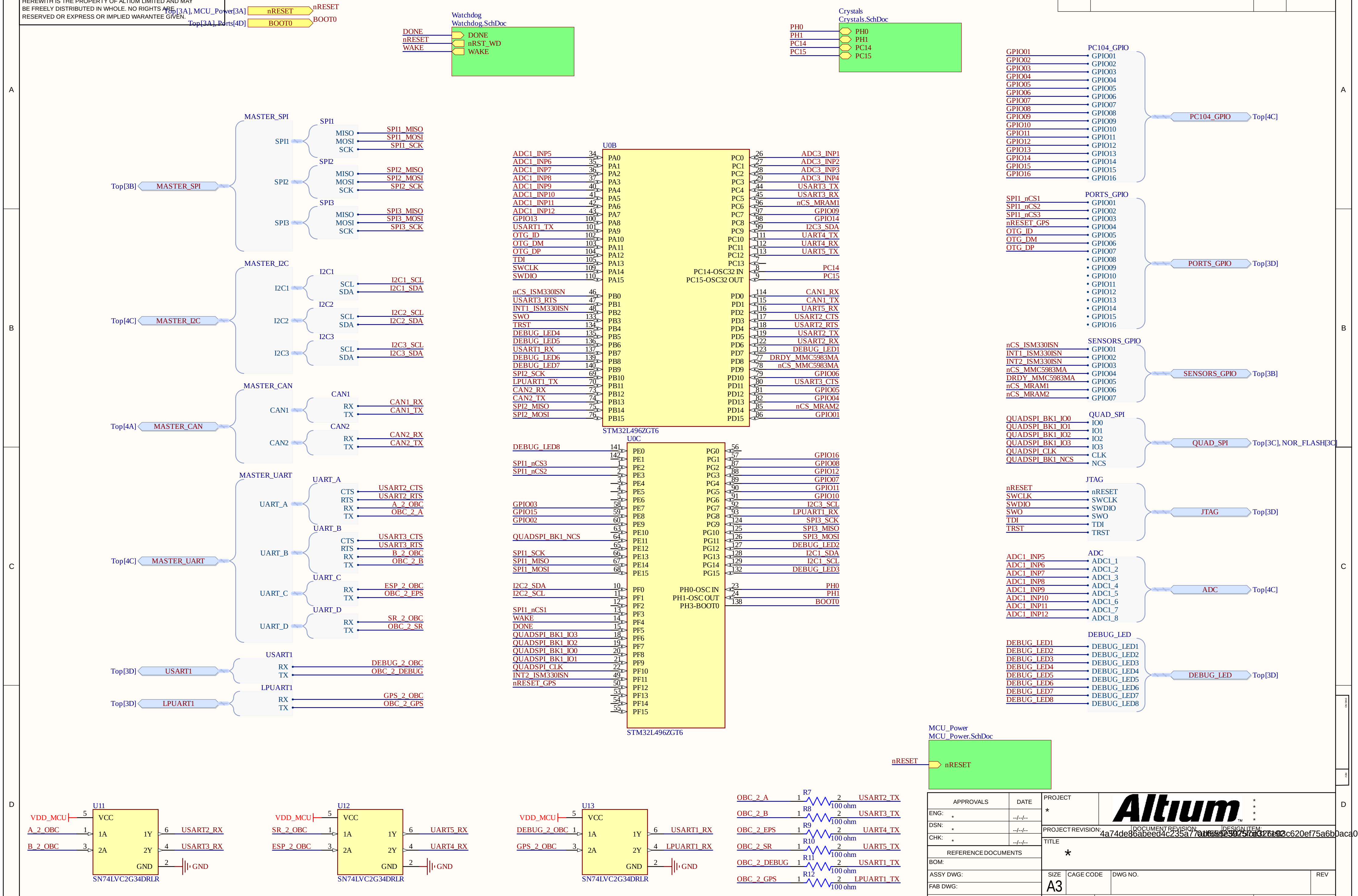
FILE NAME MAG_MMC5983MA.SchDoc
SHEET 5 OF 13

REVISION	DESCRIPTION	DATE	APPROVED



Top[3A], MCU_Power[3A]
nRESET
Top[3A], Ports[4D]
BOOT0
Watchdog
Watchdog.SchDoc
DONE
nRESET
WAKE
nRST_WD
Crystals
Crystals.SchDoc
PH0
PH1
PC14
PC15
MASTER_SPI
SPI1
SPI2
SPI3
MISO
MOSI
SCK
SPI1_MISO
SPI1_MOSI
SPI1_SCK
SPI2_MISO
SPI2_MOSI
SPI2_SCK
SPI3_MISO
SPI3_MOSI
SPI3_SCK
Top[3B]
MASTER_I2C
I2C1
I2C2
I2C3
SCL
SDA
I2C1_SCL
I2C1_SDA
I2C2_SCL
I2C2_SDA
I2C3_SCL
I2C3_SDA
Top[4C]
MASTER_CAN
CAN1
CAN2
RX
TX
CAN1_RX
CAN1_TX
CAN2_RX
CAN2_TX
Top[4A]
MASTER_UART
UART_A
UART_B
UART_C
UART_D
CTS
RTS
USART2_CTS
USART2_RTS
A_2_OBC
OBC_2_A
USART3_CTS
USART3_RTS
B_2_OBC
OBC_2_B
ESP_2_OBC
OBC_2_EPS
SR_2_OBC
OBC_2_SR
USART1
DEBUG_2_OBC
OBC_2_DEBUG
Top[3D]
LPUART1
GPS_2_OBC
OBC_2_GPS
U0B
ADC1_INP5 34 PA0
ADC1_INP6 35 PA1
ADC1_INP7 36 PA2
ADC1_INP8 37 PA3
ADC1_INP9 40 PA4
ADC1_INP10 41 PA5
ADC1_INP11 42 PA6
ADC1_INP12 43 PA7
GPIO13 100 PA8
USART1_TX 101 PA9
OTG_ID 102 PA10
OTG_DM 103 PA11
OTG_DP 104 PA12
TDI 105 PA13
SWCLK 109 PA14
SWDIO 110 PA15
nCS_ISM330ISN 46 PB0
USART3_RTS 47 PB1
INT1_ISM330ISN 48 PB2
SWO 133 PB3
TRST 134 PB4
DEBUG_LED4 135 PB5
DEBUG_LED5 136 PB6
USART1_RX 137 PB7
DEBUG_LED6 139 PB8
DEBUG_LED7 140 PB9
SPI2_SCK 69 PB10
LPUART1_TX 70 PB11
CAN2_RX 73 PB12
CAN2_TX 74 PB13
SPI2_MISO 75 PB14
SPI2_MOSI 76 PB15
PC0 26 ADC3_INP1
PC1 27 ADC3_INP2
PC2 28 ADC3_INP3
PC3 29 ADC3_INP4
PC4 44 USART3_TX
PC5 45 USART3_RX
PC6 96 nCS_MRAM1
PC7 97 GPIO09
PC8 98 GPIO14
PC9 99 I2C3_SDA
PC10 111 UART4_TX
PC11 112 UART4_RX
PC12 113 UART5_TX
PC13 7
PC14-OSC32 IN 8 PC14
PC15-OSC32 OUT 9 PC15
PD0 114 CAN1_RX
PD1 115 CAN1_TX
PD2 116 UART5_RX
PD3 117 USART2_CTS
PD4 118 USART2_RTS
PD5 119 USART2_TX
PD6 122 USART2_RX
PD7 123 DEBUG_LED1
PD8 77 DRDY_MMC5983MA
PD9 78 nCS_MMC5983MA
PD10 79 GPIO06
PD11 80 USART3_CTS
PD12 81 GPIO05
PD13 82 GPIO04
PD14 85 nCS_MRAM2
PD15 86 GPIO01
STM32L496ZGT6
U0C
DEBUG_LED8 141 PE0
142 PE1
SPI1_nCS3 1 PE2
SPI1_nCS2 2 PE3
3 PE4
4 PE5
5 PE6
GPIO03 58 PE7
GPIO15 59 PE8
GPIO02 60 PE9
63 PE10
QUADSPI_BK1_NCS 64 PE11
65 PE12
SPI1_SCK 66 PE13
SPI1_MISO 67 PE14
SPI1_MOSI 68 PE15
I2C2_SDA 10 PF0
I2C2_SCL 11 PF1
12 PF2
SPI1_nCS1 13 PF3
WAKE 14 PF4
DONE 15 PF5
QUADSPI_BK1_IO3 18 PF6
QUADSPI_BK1_IO2 19 PF7
QUADSPI_BK1_IO0 20 PF8
QUADSPI_BK1_IO1 21 PF9
QUADSPI_CLK 22 PF10
INT2_ISM330ISN 49 PF11
nRESET_GPS 50 PF12
53 PF13
54 PF14
55 PF15
PG0 56
PG1 57 GPIO16
PG2 87 GPIO08
PG3 88 GPIO12
PG4 89 GPIO07
PG5 90 GPIO11
PG6 91 GPIO10
PG7 92 I2C3_SCL
PG8 93 LPUART1_RX
PG9 124 SPI3_SCK
PG10 125 SPI3_MISO
PG11 126 SPI3_MOSI
PG12 127 DEBUG_LED2
PG13 128 I2C1_SDA
PG14 129 I2C1_SCL
PG15 132 DEBUG_LED3
PH0-OSC IN 23 PH0
PH1-OSC OUT 24 PH1
PH3-BOOT0 138 BOOT0
STM32L496ZGT6
PC104_GPIO
GPIO01 … GPIO16
Top[4C]
PORTS_GPIO
SPI1_nCS1 GPIO01
SPI1_nCS2 GPIO02
SPI1_nCS3 GPIO03
nRESET_GPS GPIO04
OTG_ID GPIO05
OTG_DM GPIO06
OTG_DP GPIO07
GPIO08 … GPIO16
Top[3D]
SENSORS_GPIO
nCS_ISM330ISN GPIO01
INT1_ISM330ISN GPIO02
INT2_ISM330ISN GPIO03
nCS_MMC5983MA GPIO04
DRDY_MMC5983MA GPIO05
nCS_MRAM1 GPIO06
nCS_MRAM2 GPIO07
Top[3B]
QUAD_SPI
QUADSPI_BK1_IO0 IO0
QUADSPI_BK1_IO1 IO1
QUADSPI_BK1_IO2 IO2
QUADSPI_BK1_IO3 IO3
QUADSPI_CLK CLK
QUADSPI_BK1_NCS NCS
Top[3C], NOR_FLASH[3C]
JTAG
nRESET
SWCLK
SWDIO
SWO
TDI
TRST
Top[3D]
ADC
ADC1_INP5 ADC1_1
ADC1_INP6 ADC1_2
ADC1_INP7 ADC1_3
ADC1_INP8 ADC1_4
ADC1_INP9 ADC1_5
ADC1_INP10 ADC1_6
ADC1_INP11 ADC1_7
ADC1_INP12 ADC1_8
Top[4C]
DEBUG_LED
DEBUG_LED1 … DEBUG_LED8
Top[3D]
MCU_Power
MCU_Power.SchDoc
nRESET
U11
VDD_MCU 5 VCC
A_2_OBC 1 1A 1Y 6 USART2_RX
B_2_OBC 3 2A 2Y 4 USART3_RX
GND 2 GND
SN74LVC2G34DRLR
U12
VDD_MCU 5 VCC
SR_2_OBC 1 1A 1Y 6 UART5_RX
ESP_2_OBC 3 2A 2Y 4 UART4_RX
GND 2 GND
SN74LVC2G34DRLR
U13
VDD_MCU 5 VCC
DEBUG_2_OBC 1 1A 1Y 6 USART1_RX
GPS_2_OBC 3 2A 2Y 4 LPUART1_RX
GND 2 GND
SN74LVC2G34DRLR
OBC_2_A 1 R7 2 USART2_TX 100 ohm
OBC_2_B 1 R8 2 USART3_TX 100 ohm
OBC_2_EPS 1 R9 2 UART4_TX 100 ohm
OBC_2_SR 1 R10 2 UART5_TX 100 ohm
OBC_2_DEBUG 1 R11 2 USART1_TX 100 ohm
OBC_2_GPS 1 R12 2 LPUART1_TX 100 ohm

APPROVALS	DATE
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REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:

PROJECT
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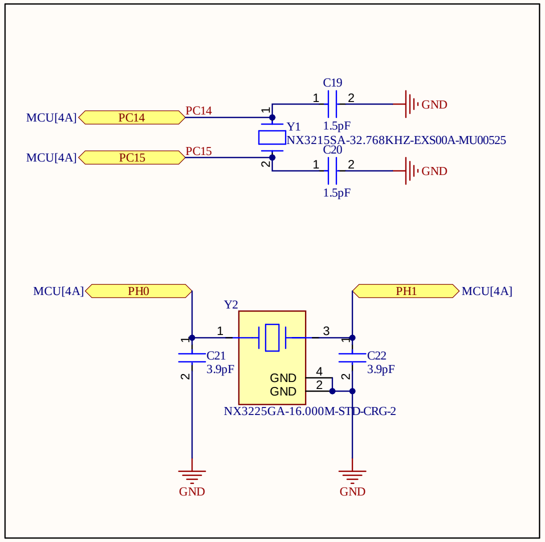
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1
2
GND
MCU[4A]
PC14
PC14
1
Y1
1.5pF
NX3215SA-32.768KHZ-EXS00A-MU00525
C20
MCU[4A]
PC15
PC15
2
1
2
GND
1.5pF
MCU[4A]
PH0
PH1
MCU[4A]
Y2
1
3
1
1
C21
3.9pF
C22
3.9pF
2
GND
GND
4
2
2
NX3225GA-16.000M-STD-CRG-2
GND
GND

APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	

REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:

PROJECT
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DESIGN ITEM: [illegible]d500a6a357f9c92bd6

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DWG NO.
REV
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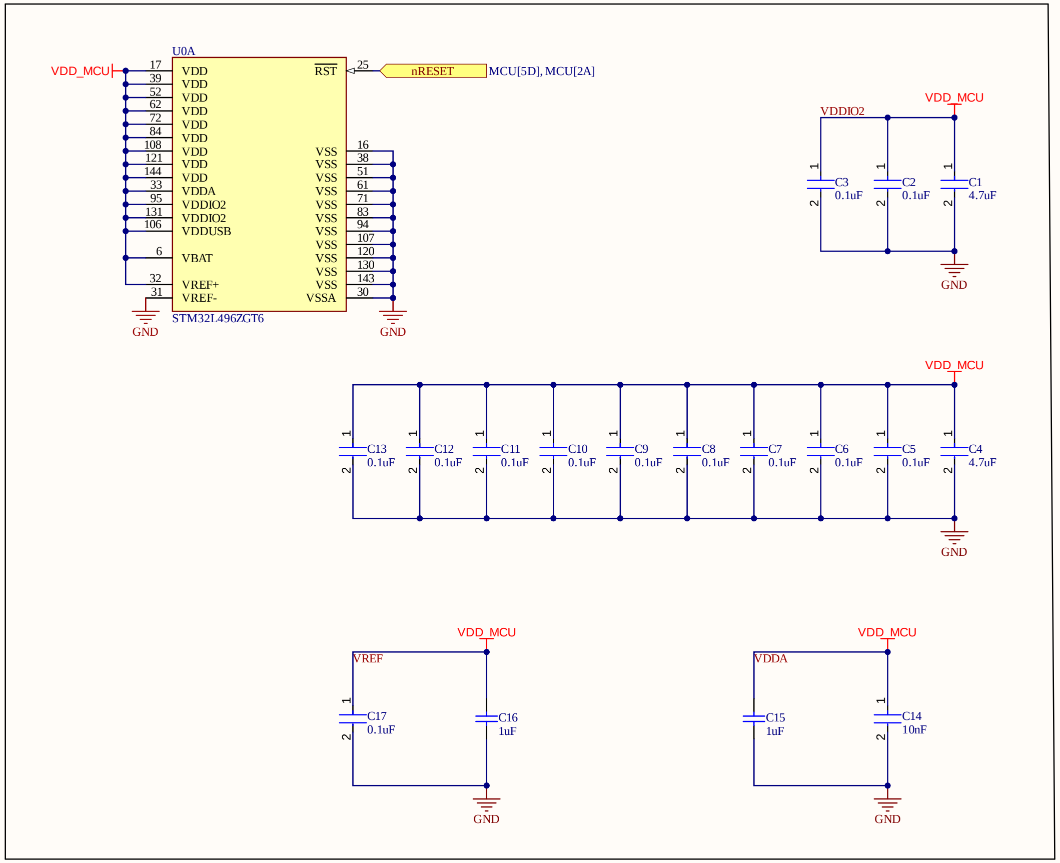
FILE NAME Crystals.SchDoc
SHEET 7 OF 13

REVISION	DESCRIPTION	DATE	APPROVED


U0A
VDD_MCU
17 39 52 62 72 84 108 121 144 33 95 131 106 6 32 31
VDD
VDD
VDD
VDD
VDD
VDD
VDD
VDD
VDD
VDDA
VDDIO2
VDDIO2
VDDUSB
VBAT
VREF+
VREF-
RST
25
nRESET
MCU[5D], MCU[2A]
16 38 51 61 71 83 94 107 120 130 143 30
VSS
VSSA
STM32L496ZGT6
GND
GND
VDDIO2
VDD_MCU
C3 0.1uF
C2 0.1uF
C1 4.7uF
GND
VDD_MCU
C13 0.1uF
C12 0.1uF
C11 0.1uF
C10 0.1uF
C9 0.1uF
C8 0.1uF
C7 0.1uF
C6 0.1uF
C5 0.1uF
C4 4.7uF
GND
VDD_MCU
VREF
C17 0.1uF
C16 1uF
GND
VDD_MCU
VDDA
C15 1uF
C14 10nF
GND
1
2

APPROVALS	DATE
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DSN: *	
CHK: *	

REFERENCE DOCUMENTS

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ASSY DWG:

FAB DWG:

PCB DWG:

PROJECT

Altium™

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PROJECT REVISION: 4a74de86abeed4c235a77[illegible]

DOCUMENT REVISION: [illegible]

DESIGN ITEM: [illegible]d500a6a357f9c92bd6

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A3			

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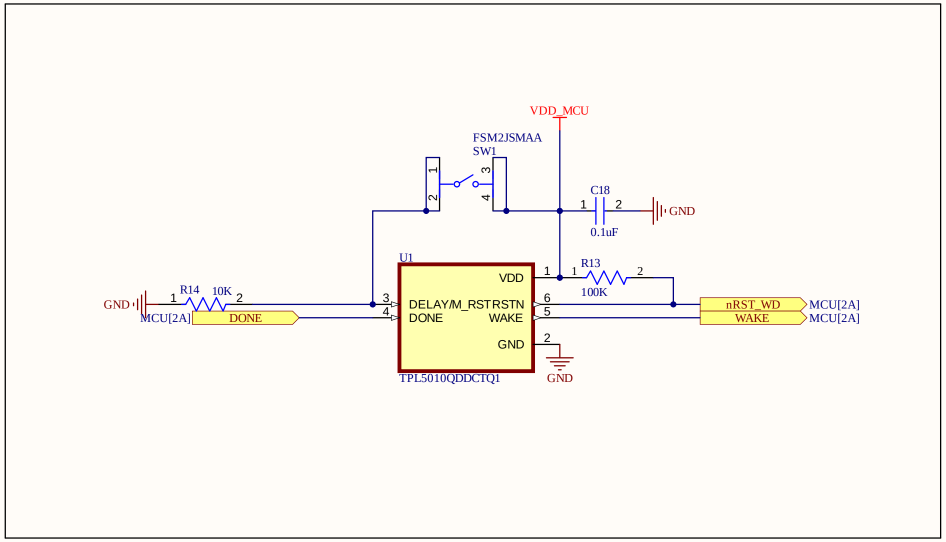
FILE NAME MCU_Power.SchDoc

SHEET 8 OF 13

REVISION	DESCRIPTION	DATE	APPROVED


VDD_MCU
FSM2JSMAA
SW1
1
2
3
4
C18
1
2
0.1uF
GND
U1
R13
1
2
100K
GND
R14
10K
1
2
MCU[2A]
DONE
3
4
DELAY/M_RST
DONE
VDD
RSTN
WAKE
GND
1
6
5
2
nRST_WD
WAKE
MCU[2A]
MCU[2A]
TPL5010QDDCTQ1
GND

APPROVALS	DATE
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CHK: *	

REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:

PROJECT Altium™ * * * *

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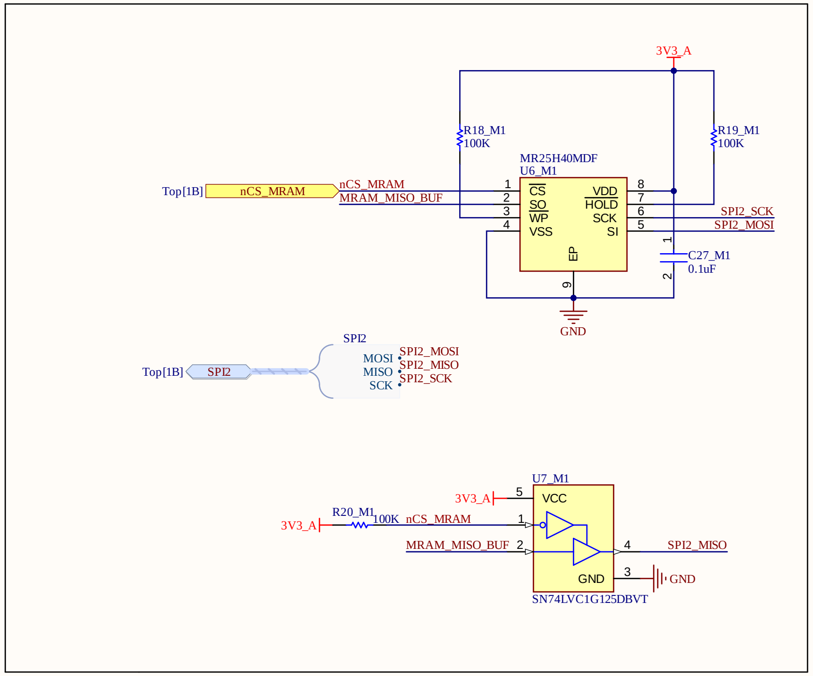
REVISION	DESCRIPTION	DATE	APPROVED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R18_M1
100K
R19_M1
100K
MR25H40MDF
U6_M1
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nCS_MRAM
MRAM_MISO_BUF
1
2
3
4
CS
SO
WP
VSS
VDD
HOLD
SCK
SI
8
7
6
5
SPI2_SCK
SPI2_MOSI
EP
9
1
2
C27_M1
0.1uF
GND
SPI2
Top[1B]
SPI2
MOSI
MISO
SCK
SPI2_MOSI
SPI2_MISO
SPI2_SCK
U7_M1
3V3_A
5
VCC
3V3_A
R20_M1
100K
nCS_MRAM
1
MRAM_MISO_BUF
2
4
SPI2_MISO
GND
3
GND
SN74LVC1G125DBVT

APPROVALS	DATE
ENG: *	
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CHK: *	

REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:

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DOCUMENT REVISION:
DESIGN ITEM:

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DWG NO.
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FILE NAME MRAM_MR25H40.SchDoc
SHEET 10 OF 13

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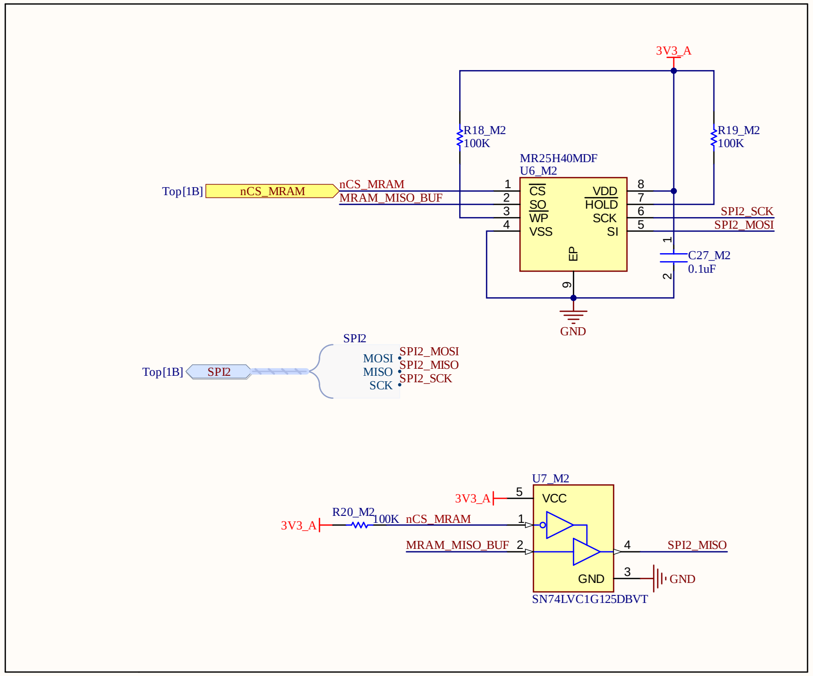
3V3_A
R18_M2
100K
R19_M2
100K
MR25H40MDF
U6_M2
Top[1B]
nCS_MRAM
nCS_MRAM
MRAM_MISO_BUF
1
2
3
4
CS
SO
WP
VSS
VDD
HOLD
SCK
SI
8
7
6
5
SPI2_SCK
SPI2_MOSI
EP
9
1
2
C27_M2
0.1uF
GND
SPI2
Top[1B]
SPI2
MOSI
MISO
SCK
SPI2_MOSI
SPI2_MISO
SPI2_SCK
U7_M2
3V3_A
5
VCC
3V3_A
R20_M2
100K
nCS_MRAM
1
MRAM_MISO_BUF
2
4
SPI2_MISO
3
GND
GND
SN74LVC1G125DBVT

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SCALE:
FILE NAME MRAM_MR25H40.SchDoc
SHEET 10 OF 13

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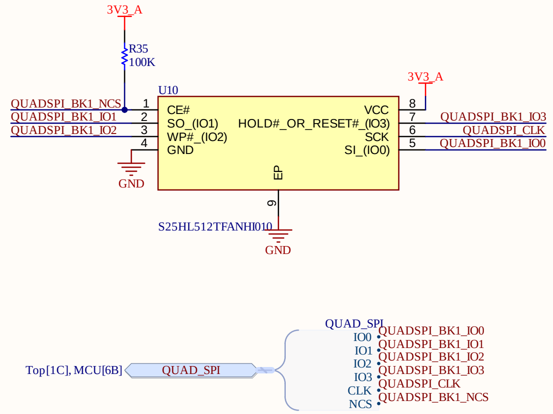
3V3_A
R35
100K
U10
QUADSPI_BK1_NCS
QUADSPI_BK1_IO1
QUADSPI_BK1_IO2
1
2
3
4
CE#
SO_(IO1)
WP#_(IO2)
GND
GND
3V3_A
8
7
6
5
VCC
HOLD#_OR_RESET#_(IO3)
SCK
SI_(IO0)
QUADSPI_BK1_IO3
QUADSPI_CLK
QUADSPI_BK1_IO0
EP
9
S25HL512TFANHI010
GND
Top[1C], MCU[6B]
QUAD_SPI
QUAD_SPI
IO0
IO1
IO2
IO3
CLK
NCS
QUADSPI_BK1_IO0
QUADSPI_BK1_IO1
QUADSPI_BK1_IO2
QUADSPI_BK1_IO3
QUADSPI_CLK
QUADSPI_BK1_NCS

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REV
SCALE:
FILE NAME NOR_FLASH.SchDoc
SHEET 11 OF 13

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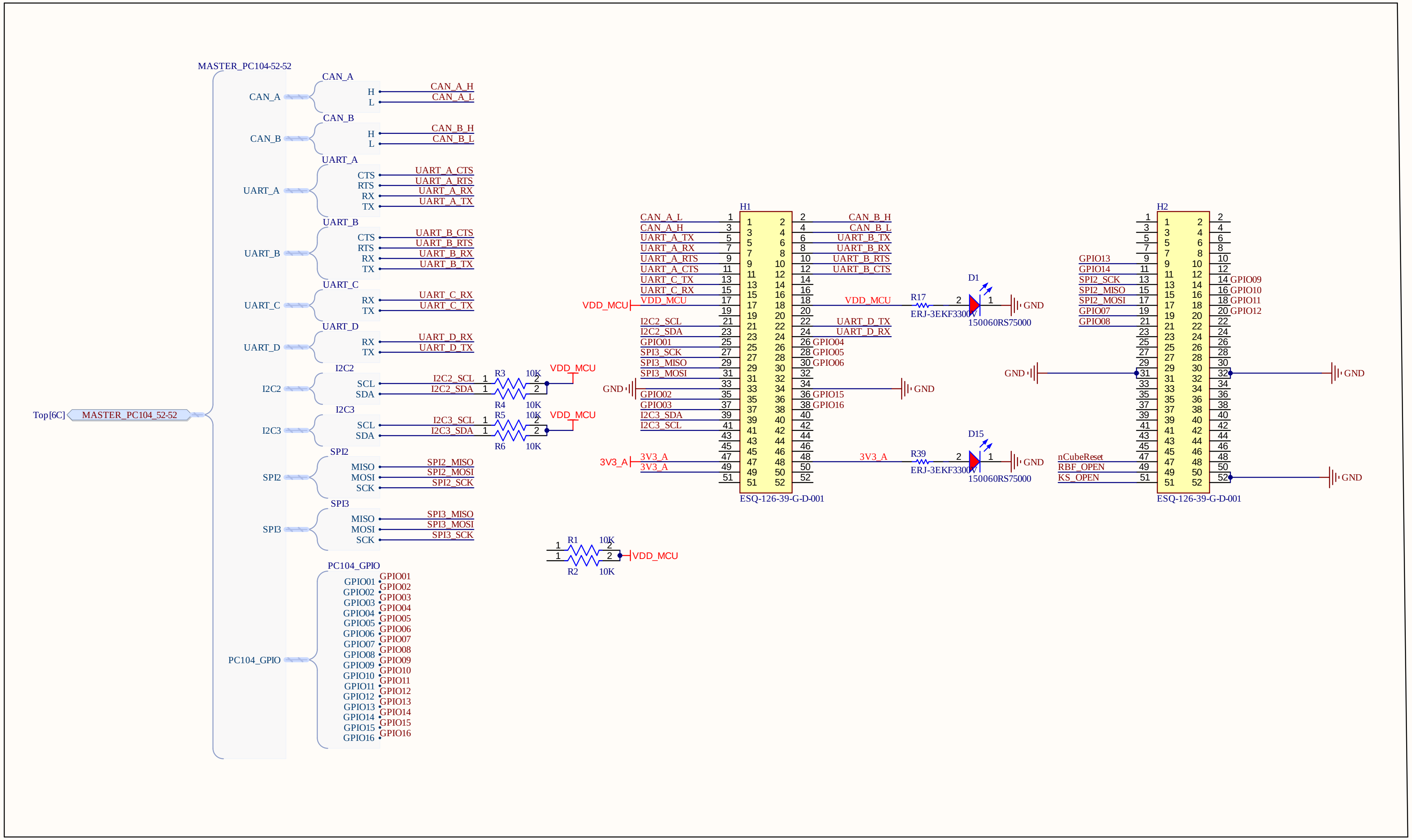
MASTER_PC104-52-52
CAN_A
H
L
CAN_A_H
CAN_A_L
CAN_B
CAN_B_H
CAN_B_L
UART_A
CTS
RTS
RX
TX
UART_A_CTS
UART_A_RTS
UART_A_RX
UART_A_TX
UART_B
UART_B_CTS
UART_B_RTS
UART_B_RX
UART_B_TX
UART_C
UART_C_RX
UART_C_TX
UART_D
UART_D_RX
UART_D_TX
I2C2
SCL
SDA
I2C2_SCL
I2C2_SDA
R3
R4
10K
VDD_MCU
I2C3
I2C3_SCL
I2C3_SDA
R5
R6
SPI2
MISO
MOSI
SCK
SPI2_MISO
SPI2_MOSI
SPI2_SCK
SPI3
SPI3_MISO
SPI3_MOSI
SPI3_SCK
PC104_GPIO
GPIO01
GPIO02
GPIO03
GPIO04
GPIO05
GPIO06
GPIO07
GPIO08
GPIO09
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
Top[6C]
MASTER_PC104_52-52
R1
R2
H1
ESQ-126-39-G-D-001
GND
3V3_A
R17
ERJ-3EKF3300V
D1
150060RS75000
R39
D15
H2
nCubeReset
RBF_OPEN
KS_OPEN

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ENG: *	
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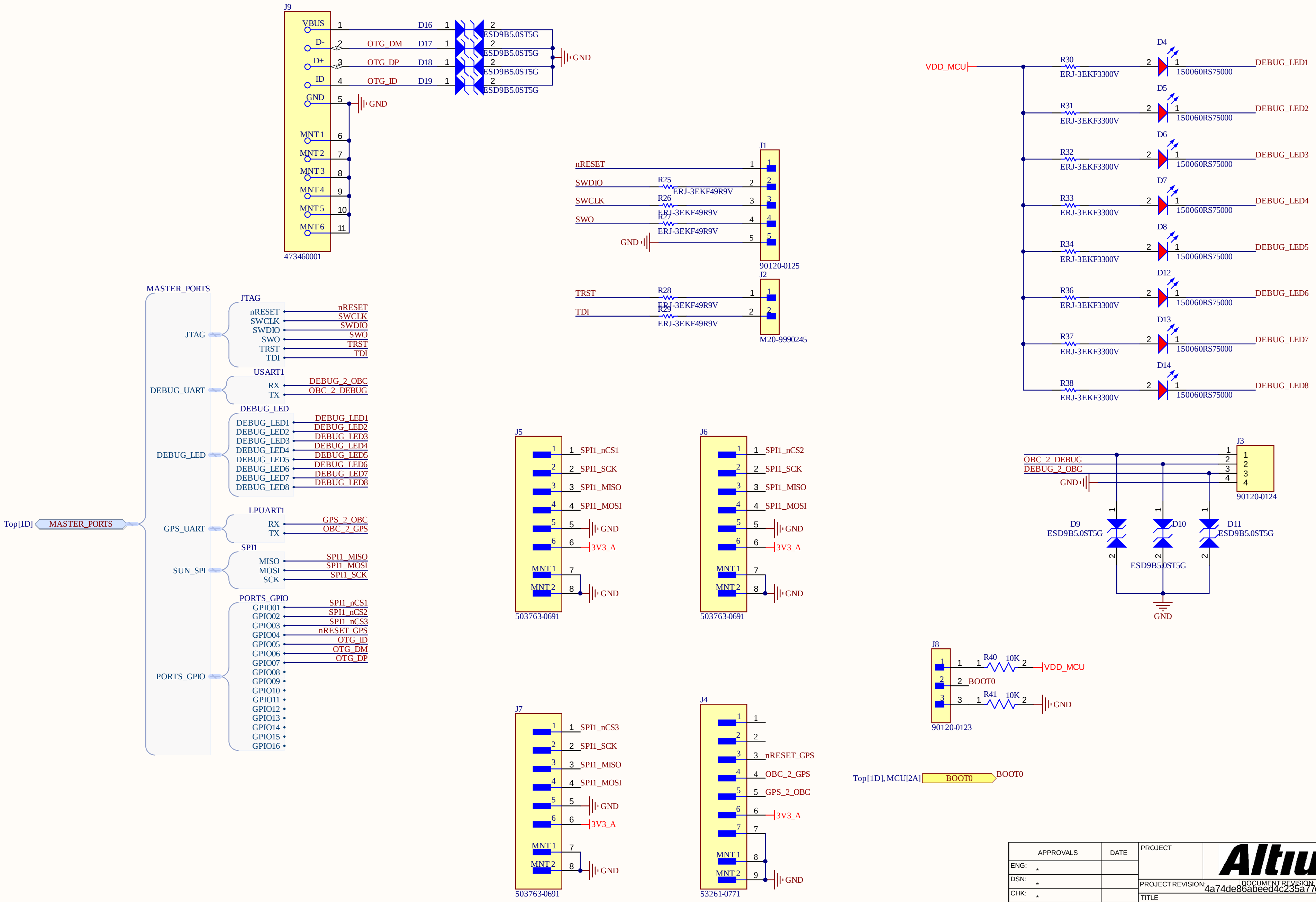
SCALE:
FILE NAME PC104_52-52.SchDoc
SHEET 12 OF 13

REVISION	DESCRIPTION	DATE	APPROVED


J9
VBUS 1
D- 2
D+ 3
ID 4
GND 5
MNT 1 6
MNT 2 7
MNT 3 8
MNT 4 9
MNT 5 10
MNT 6 11
473460001
GND
OTG_DM
OTG_DP
OTG_ID
D16 ESD9B5.0ST5G
D17 ESD9B5.0ST5G
D18 ESD9B5.0ST5G
D19 ESD9B5.0ST5G
GND
J1
nRESET
SWDIO
SWCLK
SWO
GND
R25 ERJ-3EKF49R9V
R26 ERJ-3EKF49R9V
R27 ERJ-3EKF49R9V
90120-0125
J2
TRST
TDI
R28 ERJ-3EKF49R9V
R29 ERJ-3EKF49R9V
M20-9990245
VDD_MCU
R30 ERJ-3EKF3300V
R31 ERJ-3EKF3300V
R32 ERJ-3EKF3300V
R33 ERJ-3EKF3300V
R34 ERJ-3EKF3300V
R36 ERJ-3EKF3300V
R37 ERJ-3EKF3300V
R38 ERJ-3EKF3300V
D4 150060RS75000 DEBUG_LED1
D5 150060RS75000 DEBUG_LED2
D6 150060RS75000 DEBUG_LED3
D7 150060RS75000 DEBUG_LED4
D8 150060RS75000 DEBUG_LED5
D12 150060RS75000 DEBUG_LED6
D13 150060RS75000 DEBUG_LED7
D14 150060RS75000 DEBUG_LED8
MASTER_PORTS
JTAG
nRESET
SWCLK
SWDIO
SWO
TRST
TDI
DEBUG_UART
USART1
RX DEBUG_2_OBC
TX OBC_2_DEBUG
DEBUG_LED
DEBUG_LED1
DEBUG_LED2
DEBUG_LED3
DEBUG_LED4
DEBUG_LED5
DEBUG_LED6
DEBUG_LED7
DEBUG_LED8
Top[1D] MASTER_PORTS
GPS_UART
LPUART1
RX GPS_2_OBC
TX OBC_2_GPS
SUN_SPI
SPI1
MISO SPI1_MISO
MOSI SPI1_MOSI
SCK SPI1_SCK
PORTS_GPIO
GPIO01 SPI1_nCS1
GPIO02 SPI1_nCS2
GPIO03 SPI1_nCS3
GPIO04 nRESET_GPS
GPIO05 OTG_ID
GPIO06 OTG_DM
GPIO07 OTG_DP
GPIO08
GPIO09
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
J5
1 SPI1_nCS1
2 SPI1_SCK
3 SPI1_MISO
4 SPI1_MOSI
5 GND
6 3V3_A
MNT 1 7
MNT 2 8
GND
503763-0691
J6
1 SPI1_nCS2
2 SPI1_SCK
3 SPI1_MISO
4 SPI1_MOSI
5 GND
6 3V3_A
MNT 1 7
MNT 2 8
GND
503763-0691
J7
1 SPI1_nCS3
2 SPI1_SCK
3 SPI1_MISO
4 SPI1_MOSI
5 GND
6 3V3_A
MNT 1 7
MNT 2 8
GND
503763-0691
J4
1
2
3 nRESET_GPS
4 OBC_2_GPS
5 GPS_2_OBC
6 3V3_A
7
MNT 1 8
MNT 2 9
GND
53261-0771
J3
OBC_2_DEBUG
DEBUG_2_OBC
GND
1 2 3 4
90120-0124
D9 ESD9B5.0ST5G
D10 ESD9B5.0ST5G
D11 ESD9B5.0ST5G
GND
J8
R40 10K VDD_MCU
2 BOOT0
R41 10K GND
90120-0123
Top[1D], MCU[2A] BOOT0
BOOT0

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PCB DWG:

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SHEET 13 OF 13

Board Stack Report